

CISC 667

Introduction to Autonomous Driving

Lecture 2
Sept 8, 2025

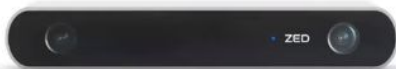
Today's Outline

- Perception sensors on an autonomous vehicle.
- Basics of camera computer vision.

Perception Sensors on an AV



RGB-D Camera



Stereo Camera



Monocular Camera



Radar



Ultrasonic



Planar LIDAR



3D LIDAR



Solid State LIDAR

Basics of Computer Vision



Erosion



Original
Image



Dilation

Basics of Computer Vision



Opening
(Erosion+Dilation)



Original
Image



Closing
(Dilation+Erosion)

Edge Detection

- Sobel Filter
- Lane Detection

Sobel Filter

- How to detect horizontal and vertical edges?

Lane Detection

